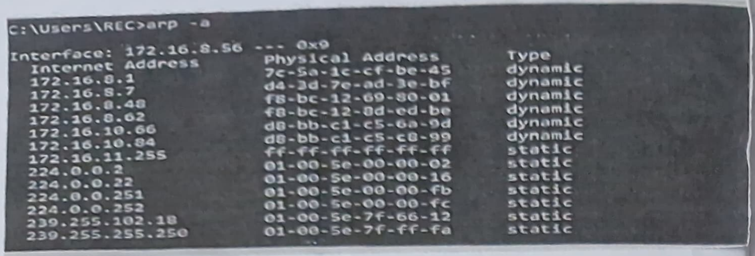
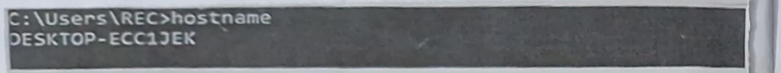
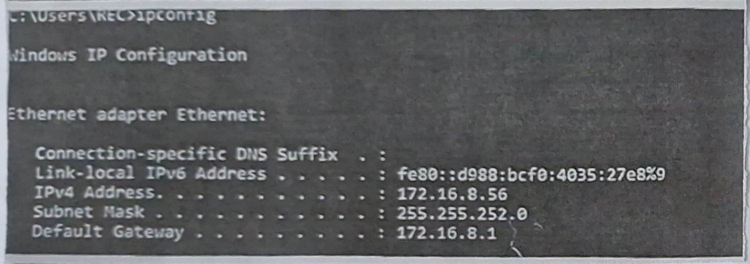


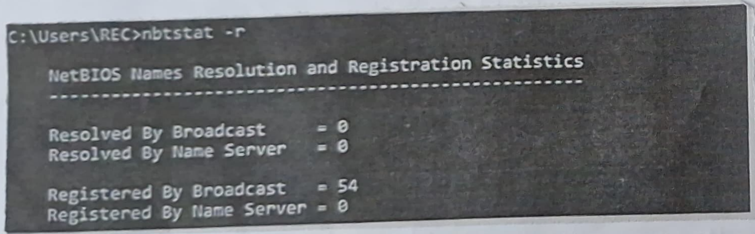
```
C:\Users\REC>arp -a

Interface: 172.16.8.56 --- 0x9
  Internet Address      Physical Address      Type
  172.16.8.1            7c-5a-1c-cf-be-45     dynamic
  172.16.8.7            d4-3d-7e-ad-3e-bf     dynamic
  172.16.8.48           f8-bc-12-69-80-01     dynamic
  172.16.8.62           f8-bc-12-8d-ed-be     dynamic
  172.16.10.66          d8-bb-c1-c5-6a-9d     dynamic
  172.16.10.84          d8-bb-c1-c5-c8-99     dynamic
  172.16.11.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```



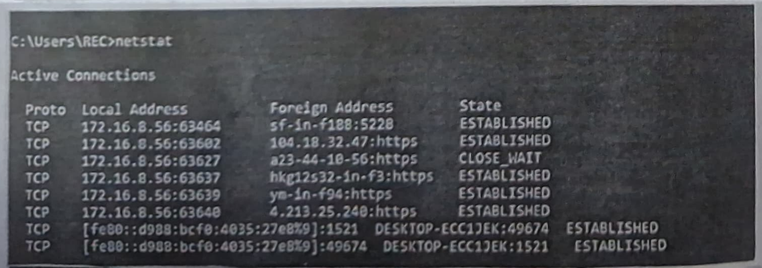
```
C:\Users\REC>hostname
DESKTOP-ECC1JEK
```



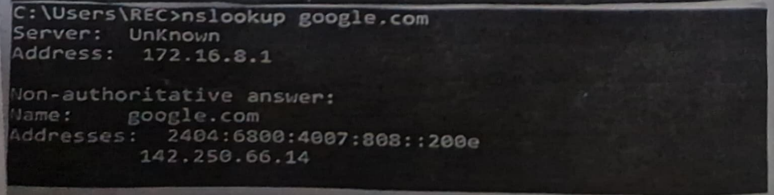
```
C:\Users\REC>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d988:bcf0:4035:27e8%9
   IPv4 Address. . . . . . . . . . . : 172.16.8.56
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1
```



```
C:\Users\REC>nbtstat -r

    NetBIOS Names Resolution and Registration Statistics
    ----------------------------------------------------

    Resolved By Broadcast     = 0
    Resolved By Name Server   = 0

    Registered By Broadcast   = 54
    Registered By Name Server = 0
```



```
C:\Users\REC>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.8.56:63464      sf-in-f188:5228        ESTABLISHED
  TCP    172.16.8.56:63602      104.18.32.47:https     ESTABLISHED
  TCP    172.16.8.56:63627      a23-44-10-56:https     CLOSE_WAIT
  TCP    172.16.8.56:63637      hkg12s32-in-f3:https   ESTABLISHED
  TCP    172.16.8.56:63639      ym-in-f94:https        ESTABLISHED
  TCP    172.16.8.56:63640      4.213.25.240:https     ESTABLISHED
  TCP    [fe80::d988:bcf0:4035:27e8%9]:1521  DESKTOP-ECC1JEK:49674  ESTABLISHED
  TCP    [fe80::d988:bcf0:4035:27e8%9]:49674  DESKTOP-ECC1JEK:1521   ESTABLISHED
```



```
C:\Users\REC>nslookup google.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:808::200e
          142.250.66.14
```

EXPERIMENT - 1

Aim:

Study of various networks commands used in Linux and windows.

Basic Networking commands:

arp -a.

ARP is a short form of address resolution protocol. It will show the IP address of your computer along with the IP addres and MAC.

host name:

This is the simplest of all TCP/IP commands. It displays name of your computer.

ipconfig /all:

This command display detailed configuration information about your TCP/IP connection including router gateway, DNS, DHCP and type of ethernet adapter.

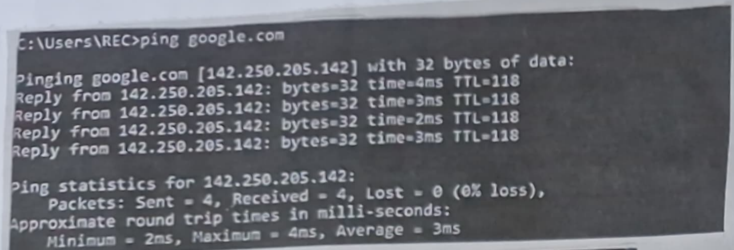
nbtstat -a:

This command helps solve problem with Net Bios name resolution. It stands for netBIOS over TCP/IP

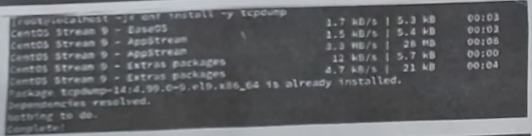
netstat:

It displays a variety of statistics about a computer active TCP/IP connections. It is a command line tool for monitoring network connections.

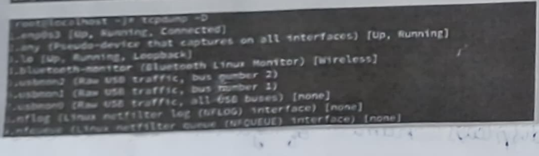
eg; netstat -r.



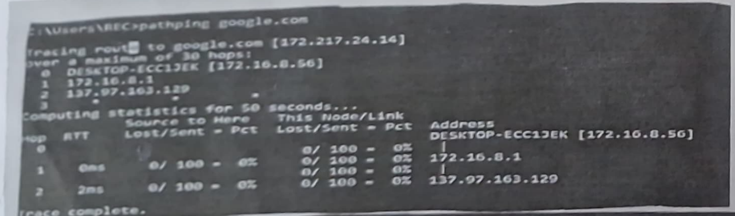
```
C:\Users\REC>ping google.com

Pinging google.com [142.250.205.142] with 32 bytes of data:
Reply from 142.250.205.142: bytes=32 time=4ms TTL=118
Reply from 142.250.205.142: bytes=32 time=3ms TTL=118
Reply from 142.250.205.142: bytes=32 time=2ms TTL=118
Reply from 142.250.205.142: bytes=32 time=3ms TTL=118

Ping statistics for 142.250.205.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```



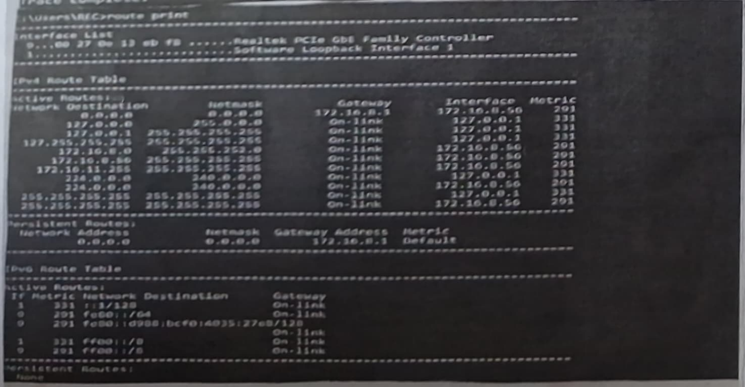
```
[root@localhost ~]# dnf install -y tcpdump
CentOS Stream 9 - BaseOS                        1.7 kB/s | 5.3 kB     00:03
CentOS Stream 9 - BaseOS                        1.5 kB/s | 5.4 kB     00:03
CentOS Stream 9 - AppStream                     3.3 MB/s |  28 MB     00:08
CentOS Stream 9 - AppStream                      12 kB/s | 5.7 kB     00:00
CentOS Stream 9 - Extras packages               4.7 kB/s |  21 kB     00:04
Package tcpdump-14:4.99.0-9.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```



```
[root@localhost ~]# tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]
8.nflog (Linux netfilter log (NFLOG) interface) [none]
9.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
```



```
C:\Users\REC>pathping google.com

Tracing route to google.com [172.217.24.14]
over a maximum of 30 hops:
  0  DESKTOP-ECC1JEK [172.16.8.56]
  1  172.16.8.1
  2  137.97.163.129
  3     *        *        *
Computing statistics for 50 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-ECC1JEK [172.16.8.56]
                                0/ 100 =  0%   |
  1    0ms     0/ 100 =  0%     0/ 100 =  0%  172.16.8.1
                                0/ 100 =  0%   |
  2    2ms     0/ 100 =  0%     0/ 100 =  0%  137.97.163.129

Trace complete.
```



```
C:\Users\REC>route print
===========================================================================
Interface List
  9...00 27 0e 13 eb f8 ......Realtek PCIe GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0       172.16.8.1      172.16.8.56    291
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
       172.16.8.0    255.255.252.0         On-link       172.16.8.56    291
      172.16.8.56  255.255.255.255         On-link       172.16.8.56    291
    172.16.11.255  255.255.255.255         On-link       172.16.8.56    291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link       172.16.8.56    291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link       172.16.8.56    291
===========================================================================
Persistent Routes:
  Network Address          Netmask  Gateway Address  Metric
          0.0.0.0          0.0.0.0       172.16.8.1  Default
===========================================================================

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  9    291 fe80::/64                On-link
  9    291 fe80::d988:bcf0:4035:27e8/128
                                    On-link
  1    331 ff00::/8                 On-link
  9    291 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

ns lookup :

It is a tool used to perform DNS lookups in linux. It is used to display DNS details, ins records for a domain. It can operate in two modes: ~~inhatance~~ interactive & non interactive.

pathping :

It is a unique to windows and is basically a combination of the ping & tracert commands.

ping :

It is the best way to test connectivity b/w two nodes.

① # ping hostname (ping localhost)

② # ping ip address (ping 4.2.2:2

③ # ping fully qualified domain name.

Route :

route command is used to show the IP routing table. It is primarily used to setup static routes to specific host or networks.

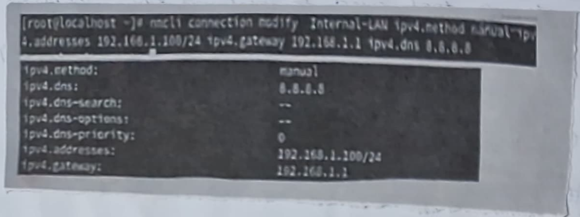
Some important Linux networking commands:

① ip :

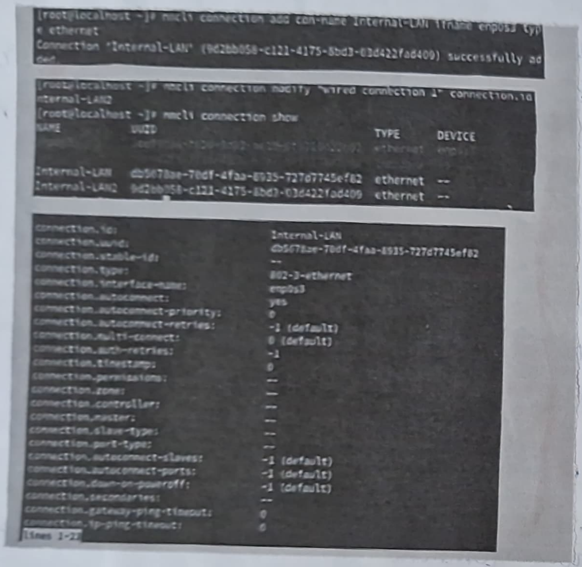
It is one of the basic commands every admin will need daily work from setting up new systems and assigning ips troubleshooting existing systems.

syntax: ip <options> <OBJECT> <COMMAND>

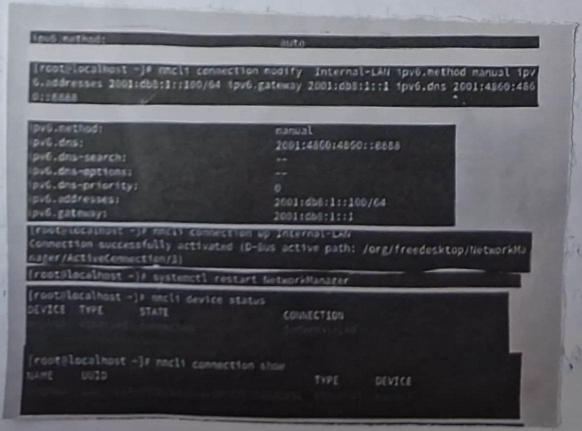
eg: [root@server ~]# ip address show.



```
[root@localhost ~]# nmcli connection modify  Internal-LAN ipv4.method manual ipv
4.addresses 192.168.1.100/24 ipv4.gateway 192.168.1.1 ipv4.dns 8.8.8.8

ipv4.method:                              manual
ipv4.dns:                                 8.8.8.8
ipv4.dns-search:                          --
ipv4.dns-options:                         --
ipv4.dns-priority:                        0
ipv4.addresses:                           192.168.1.100/24
ipv4.gateway:                             192.168.1.1
```



```
[root@localhost ~]# nmcli connection add con-name Internal-LAN ifname enp0s3 typ
e ethernet
Connection "Internal-LAN" (9d2bb058-c121-4175-8bd3-03d422fad409) successfully ad
ded.

[root@localhost ~]# nmcli connection modify "Wired connection 1" connection.id
nternal-LAN2
[root@localhost ~]# nmcli connection show
NAME          UUID                                  TYPE      DEVICE

Internal-LAN  db5678ae-70df-4faa-8935-727d7745ef82  ethernet  --
Internal-LAN2 9d2bb058-c121-4175-8bd3-03d422fad409  ethernet  --

connection.id:                            Internal-LAN
connection.uuid:                          db5678ae-70df-4faa-8935-727d7745ef82
connection.stable-id:                     --
connection.type:                          802-3-ethernet
connection.interface-name:                enp0s3
connection.autoconnect:                   yes
connection.autoconnect-priority:          0
connection.autoconnect-retries:           -1 (default)
connection.multi-connect:                 0 (default)
connection.auth-retries:                  -1
connection.timestamp:                     0
connection.permissions:                   --
connection.zone:                          --
connection.controller:                    --
connection.master:                        --
connection.slave-type:                    --
connection.port-type:                     --
connection.autoconnect-slaves:            -1 (default)
connection.autoconnect-ports:             -1 (default)
connection.down-on-poweroff:              -1 (default)
connection.secondaries:                   --
connection.gateway-ping-timeout:          0
connection.ip-ping-timeout:               0
lines 1-27
```



```
ipv6.method:                              auto

[root@localhost ~]# nmcli connection modify  Internal-LAN ipv6.method manual ipv
6.addresses 2001:db8:1::100/64 ipv6.gateway 2001:db8:1::1 ipv6.dns 2001:4860:486
0::8888

ipv6.method:                              manual
ipv6.dns:                                 2001:4860:4860::8888
ipv6.dns-search:                          --
ipv6.dns-options:                         --
ipv6.dns-priority:                        0
ipv6.addresses:                           2001:db8:1::100/64
ipv6.gateway:                             2001:db8:1::1
[root@localhost ~]# nmcli connection up Internal-LAN
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkMa
nager/ActiveConnection/3)
[root@localhost ~]# systemctl restart NetworkManager
[root@localhost ~]# nmcli device status
DEVICE  TYPE      STATE                 CONNECTION


[root@localhost ~]# nmcli connection show
NAME    UUID                                  TYPE      DEVICE
```

② **if config :**

It is a stable in many systems andmn tool. belt for configuring and troubleshooting networks.

③ **mtr :**

It is a program with a command line interface. that serves as a network diagnostic and troubleshooting tool.

syntax : mtr <OPTIONS> hostname / IP

eg : [root @ server] # mtr. googles. com.

④ **tcp dump :**

This command is designed for capturing and displaying packets you can install tcpdum.

[root @ server] # dnf install - y tcpdump

⑤ **png :**

It is a tool that verifies IP. level. connectivity to another. TCP /IP computer. by sending Internet control Message protocol. echo request messages. The receipt of corresponding echo reples messay

[root @ server.] .#png google.com.

PING google.com (216.58.206.174) 56(84) bytes of data.

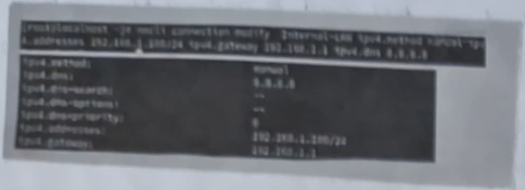
64 bytes from 1of+02827 -in f14. 1e100 .net (216.58.206.174)

seq = 1 ttl = 56 time = 10.7ms.

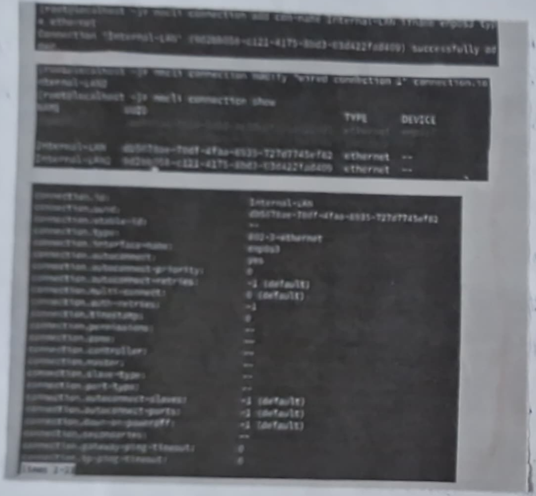
nmcli commands.

① nmcli connection show.

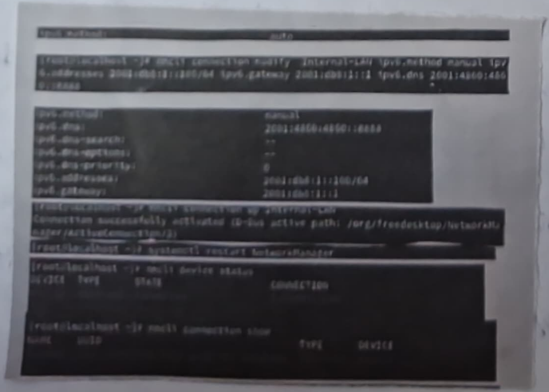
Displays all available network manager connection profiles and their associated devices



```
[root@localhost ~]# nmcli connection modify  Internal-LAN ipv4.method manual ip
4.addresses 192.168.1.100/24 ipv4.gateway 192.168.1.1 ipv4.dns 8.8.8.8
ipv4.method:                                    manual
ipv4.dns:                                       8.8.8.8
ipv4.dns-search:                                --
ipv4.dns-options:                               --
ipv4.dns-priority:                              0
ipv4.addresses:                                 192.168.1.100/24
ipv4.gateway:                                   192.168.1.1
```



```
[root@localhost ~]# nmcli connection add con-name Internal-LAN ifname enp0s3 typ
e ethernet
Connection 'Internal-LAN' (9d2b0058-c121-4175-8bd3-e3d422fa0409) successfully ad
ded.
[root@localhost ~]# nmcli connection modify "Wired connection 1" connection.id
Internal-LAN2
[root@localhost ~]# nmcli connection show
NAME           UUID                                  TYPE      DEVICE
Internal-LAN   d0507bce-70df-4fa0-8935-727d7745ef82  ethernet  --
Internal-LAN2  9d2b0058-c121-4175-8bd3-e3d422fa0409  ethernet  --
connection.id:                          Internal-LAN
connection.uuid:                        d0507bce-70df-4fa0-8935-727d7745ef82
connection.stable-id:                   --
connection.type:                        802-3-ethernet
connection.interface-name:              enp0s3
connection.autoconnect:                 yes
connection.autoconnect-priority:        0
connection.autoconnect-retries:         -1 (default)
connection.multi-connect:               0 (default)
connection.auth-retries:                -1
connection.timestamp:                   0
connection.permissions:                 --
connection.zone:                        --
connection.controller:                  --
connection.master:                      --
connection.slave-type:                  --
connection.port-type:                   --
connection.autoconnect-slaves:          -1 (default)
connection.autoconnect-ports:           -1 (default)
connection.down-on-poweroff:            -1 (default)
connection.secondaries:                 --
connection.gateway-ping-timeout:        0
connection.ip-ping-timeout:             0
lines 1-22
```



```
ipv6.method:                                    auto
[root@localhost ~]# nmcli connection modify  Internal-LAN ipv6.method manual ipv
6.addresses 2001:db8:1::100/64 ipv6.gateway 2001:db8:1::1 ipv6.dns 2001:4860:486
0::8888
ipv6.method:                                    manual
ipv6.dns:                                       2001:4860:4860::8888
ipv6.dns-search:                                --
ipv6.dns-options:                               --
ipv6.dns-priority:                              0
ipv6.addresses:                                 2001:db8:1::100/64
ipv6.gateway:                                   2001:db8:1::1
[root@localhost ~]# nmcli connection up Internal-LAN
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkMa
nager/ActiveConnection/3)
[root@localhost ~]# systemctl restart NetworkManager
[root@localhost ~]# nmcli device status
DEVICE  TYPE      STATE                          CONNECTION
[root@localhost ~]# nmcli connection show
NAME    UUID                                     TYPE      DEVICE
```

② if config:
It is a [illegible]
[illegible] for configuring and [illegible]

③ mtr:
It is a program with a [illegible]
interface. that [illegible]
and troubleshooting tool
syntax: mtr <OPTIONS> hostname/IP
eg: [root@ [illegible]] # mtr [illegible]

④ tcpdump:
This command is designed for [illegible]
displaying packets you can install [illegible]
[root@ server] # dnf install -y tcpdump

⑤ ping:
It is a tool that verifies IP level connectivity
to another TCP/IP computer. by sending [illegible]
control Message protocol. echo request message
The receipt of corresponding echo reply [illegible]
[root @ server] # ping google.com.
PING google.com (216.58.206.174) 56(84) bytes of data
64 bytes from rof+02827 -in f14.1e100.net (216.58. [illegible]
seq = 1 ttl = 56 time = 10.7 ms.

nmcli commands.

① nmcli connection show.
Displays all available network [illegible]

② nmcli connection add con-name. <connection name> if name <devicename> type ethernet created a new ethernet connection profile for the specified network interface

③ nmcli connection. modify "old connection" connection. id internal-LAN 2 modifies the settings of an. existing method

④ Auto configures the connection to automatically obtain an IPV6 address using

⑤ nmcli connection modify "internal-LAN" input method manual IPv4 address 192.0.21.124 IPV4 gateway 192.02.254.

Assigns = static IPv6 address, gateway DNS serve

⑥ nmcli connection. up. internal -LAN. ipv6 method. auto configures the connection to automatically obtain. IPv6 addr.

⑦ nmcli connection up internal-LAN activates the specified.

⑧ nmcli connection modify internal-LAN IPV6 method manual IPv6 address 2001:db8:0::100/64. IPV6 gateway 900:1068:1::9 IPv6 2001:4860:4860: 8888

Assigns a static IPv6 address gateway DNS servers to the connection

9/10

10/9

Result:

Thus the study of various network commands used in Linux and windows is done and verified